

Advanced Topics in Computer Graphics I

Volume Rendering



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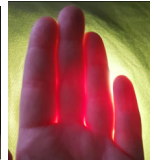
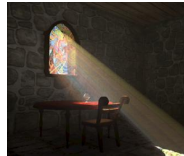


Volume scattering process



In many scenarios **scattering effects** can not be omitted. The vacuum assumption often made in global illumination simulations is too restrictive!

- ▶ spotlight through fog (god rays)
- ▶ skin
- ▶ smoke
- ▶ clouds,
- ▶ liquids
- ▶ transparent materials
- ▶ blue sky, red sunset



Rendering at Low Concentrations



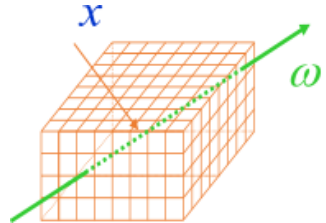
Rendering at Natural Concentrations



Emission: if light is emitted within the volume and is not absorbed or scattered while travelling through the medium the increase of radiance along a ray is proportional to the emitted radiance.

Adding a heat source g inside the volume Ω leads to

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) = \sigma_a(\boldsymbol{x})L_e(\boldsymbol{x}, \boldsymbol{\omega}),$$



where $L_e(\boldsymbol{x}, \boldsymbol{\omega})$ is the source term and $\boldsymbol{x}$ is a point along the ray in direction of $\boldsymbol{\omega}$. The proportionality coefficient $\sigma_a(\boldsymbol{x})$ is often assumed to be 1.



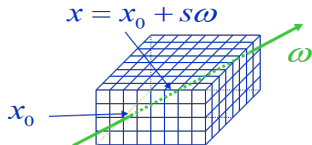
The **solution** to the differential equation

$$(\omega \cdot \nabla)L(x, \omega) = \sigma_a(x)L_e(x, \omega),$$

is given by

$$L(x_0 + s\omega, \omega) = L(x_0, \omega) + \int_0^s \sigma_a(x_0 + t\omega)L_e(x_0 + t\omega, \omega)dt$$

where $L(x_0, \omega)$ is radiance at the point x_0 . The solution describes the intensity that arrives at the point on the ray at x_0 from the direction ω and usually represents a background luminance.





Another simple model is based on the assumption that light decreases by **absorption** and **scattering** within the volume. The corresponding differential equations are

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) = -\sigma_s(\boldsymbol{x})L(\boldsymbol{x}, \boldsymbol{\omega})$$

and

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) = -\sigma_a(\boldsymbol{x})L(\boldsymbol{x}, \boldsymbol{\omega})$$

where $\sigma_s(\boldsymbol{x}) = \sigma_s(\boldsymbol{x}, \boldsymbol{\omega})$ is the scattering coefficient and $\sigma_a(\boldsymbol{x}) = \sigma_a(\boldsymbol{x}, \boldsymbol{\omega})$ is the absorption coefficient.

Combining both equations, i.e. taking both effects into account, we get

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) = -\sigma_t(\boldsymbol{x})L(\boldsymbol{x}, \boldsymbol{\omega})$$

where $\sigma_t(\boldsymbol{x}, \boldsymbol{\omega}) = (\sigma_a(\boldsymbol{x}, \boldsymbol{\omega}) + \sigma_s(\boldsymbol{x}, \boldsymbol{\omega}))$ is referred to as the **extinction coefficient** and

describes the decrease in luminance in volume. In graphic literature, this coefficient is often referred to as **opacity**. It is a measure of the impermeability of the medium.



The **solution** to this differential equation is given by the **Beer-Lambert law, 1760**

$$L(\mathbf{x}_0 + s\boldsymbol{\omega}, \boldsymbol{\omega}) = L(\mathbf{x}_0, \boldsymbol{\omega}) \cdot e^{-\tau(\mathbf{x}_0, \mathbf{x})}$$

where

$$\tau(\mathbf{x}_0, \mathbf{x} = \mathbf{x}_0 + s\boldsymbol{\omega}) = \int_0^s \sigma_t(\mathbf{x}_0 + t\boldsymbol{\omega}) dt$$

is the **optical thickness** of the material along the ray between the points $\mathbf{x}_0$ and $\mathbf{x}$.
 $L(\mathbf{x}_0, \boldsymbol{\omega})$ defines the boundary condition.

$T(\mathbf{x}_0, \mathbf{x}) := e^{-\tau(\mathbf{x}_0, \mathbf{x})}$ is called **transmittance** and gives the fraction of light that makes it from $\mathbf{x}_0$ to $\mathbf{x}$.

In a homogenous medium, σ_t is constant, so the integral can be easily computed and we get $\tau(\mathbf{x}_0, \mathbf{x} = \mathbf{x}_0 + s\boldsymbol{\omega}) = s\sigma_t$, leading to

$$L(\mathbf{x}_0 + s\boldsymbol{\omega}, \boldsymbol{\omega}) = L(\mathbf{x}_0, \boldsymbol{\omega}) \cdot e^{-s\sigma_t}$$



If we combine all effects in one differential equation we get

$$(\omega \cdot \nabla)L(x, \omega) = \sigma_a(x)L_e(x, \omega) - \sigma_t(x)L(x, \omega)$$

This is an inhomogeneous linear ordinary differential equation in the parameter s along the ray of the form

$$\underbrace{y'(s)}_{(\omega \cdot \nabla)L(x(s), \omega)} = \underbrace{b(s)}_{\sigma_a(x(s))L_e(x(s), \omega)} + \underbrace{a(s)y(s)}_{-\sigma_t(x(s))L(x(s), \omega)}$$

It can be solved using the method called **variation of constants**:



Let $\phi(s)$ be a solution of the homogenous equation,i.e.

$$\phi'(s) = a(s)\phi(s)$$

In case that $\phi(s_0) \neq 0$ it follows that $\phi(s) \neq 0, \quad \forall s$ and therefore every solution ψ of the inhomogenous equation can be written as $\psi(s) = \phi(s)u(s)$ for some function $u(s)$. This leads to the following two equations

$$\psi'(s) = a(s)\psi(s) + b(s) = a(s)\phi(s)u(s) + b(s)$$

$$\psi'(s) = \phi'(s)u(s) + \phi(s)u'(s) = a(s)\phi(s)u(s) + \phi(s)u'(s)$$

It follows that

$$u'(s) = \phi^{-1}(s)b(s)$$

and therefore

$$u(s) = \int_{s_0}^s \phi^{-1}(t)b(t)dt + c$$

where c is a constant.



The combination of the effects emission, absorbtion and out-scattering leads to

$$(\omega \cdot \nabla)L(x, \omega) = \sigma_a(x)L_e(x, \omega) - \sigma_t(x)L(x, \omega)$$

Integrating this non-homogeneous differential equation using the variation of constants method we get

$$L(x_0 + s\omega, \omega) = L(x_0, \omega) \cdot e^{-\tau(x_0, x)} + \int_0^s e^{-\tau(x_0 + t, x_0 + s\omega)} \sigma_a(x_0 + t\omega) L_e(x_0 + t\omega, \omega) dt$$

Note, this solution does not contain any increase of the radiance due to in-scattered light.



If we also consider the **in-scattering** of light at a position with parameter x along the ray an additional term must be added:

$$(\omega \cdot \nabla)L(x, \omega) = \sigma_s(x) \int_{\Omega} p(x, \omega', \omega) L_i(x, \omega') d\omega'$$

where $L_i(x, \omega')$ is the in-scattered radiance from direction ω' and $p(x, \omega', \omega)$ is the **medium phase function** describing the probability that incoming light from direction ω' is scattered at position x into the direction ω .

Combining all equations, i.e. taking all effects into account, we get the

Radiative transfer equation RTE:

$$(\omega \cdot \nabla)L(x, \omega) = \sigma_a(x)L_e(x, \omega) - \sigma_t(x)L(x, \omega) + \sigma_s(x) \int_{\Omega} p(x, \omega', \omega) L_i(x, \omega') d\omega'$$



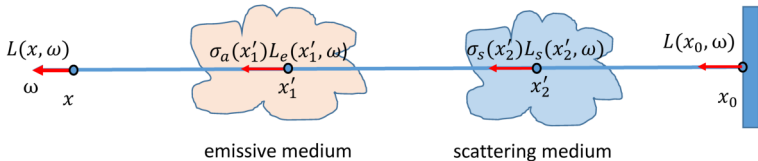
$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) = -\sigma_a(\boldsymbol{x})L_e(\boldsymbol{x}, \boldsymbol{\omega}) - \sigma_t(\boldsymbol{x})L(\boldsymbol{x}, \boldsymbol{\omega}) + \sigma_s(\boldsymbol{x}) \int_{\Omega} p(\boldsymbol{x}, \boldsymbol{\omega}', \boldsymbol{\omega})L_i(\boldsymbol{x}, \boldsymbol{\omega}')d\boldsymbol{\omega}'$$

Parametrizing $\boldsymbol{x}(t) = \boldsymbol{x}_0 + t\boldsymbol{\omega}$ along the ray and integrating the non-homogeneous radiative transfer equation in t using the variation of constants method we get

$$\begin{aligned} L(\boldsymbol{x} = \boldsymbol{x}_0 + s\boldsymbol{\omega}, \boldsymbol{\omega}) &= L(\boldsymbol{x}_0, \boldsymbol{\omega}) \cdot e^{-\tau(\boldsymbol{x}_0, \boldsymbol{x}_0 + s\boldsymbol{\omega})} \\ &+ \int_0^s e^{-\tau(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{x}_0 + s\boldsymbol{\omega})} \sigma_a(\boldsymbol{x}_0 + t\boldsymbol{\omega}) L_e(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{\omega}) dt \\ &+ \int_0^s e^{-\tau(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{x}_0 + s\boldsymbol{\omega})} \sigma_s(\boldsymbol{x}_0 + t\boldsymbol{\omega}) \\ &\quad \underbrace{\int_{\Omega} p(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{\omega}', \boldsymbol{\omega}) L_i(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{\omega}') d\boldsymbol{\omega}' dt}_{L_s(\boldsymbol{x}_0 + t\boldsymbol{\omega}, \boldsymbol{\omega})} \end{aligned}$$



Volume rendering equation



Considering the nearest surface point $x_0 = x - s\omega$ at distance s , substituting back $x' = x_0 + t\omega$ and collecting terms the resulting volume rendering equation (VRE) reads

Volume Rendering Equation:

$$\begin{aligned} L(x, \omega) &= L(x_0, \omega) \cdot e^{-\tau(x_0, x)} \\ &+ \int_0^s e^{-\tau(x', x)} [\sigma_a(x')L_e(x', \omega) + \sigma_s(x')L_s(x', \omega)] dt \end{aligned}$$



The phase function



The phase function describes the distribution of scattered light in the medium. In contrast to the BRDF, the phase function is dimensionless and normalized:

$$\int_{\Omega} p(\mathbf{x}, \omega', \omega) d\omega' = 1.$$

An important property of the phase functions is that they are **reciprocal**, i.e. incoming and outgoing direction can be interchanged without changing the function's value. In most naturally occurring media, the phase function is a 1D function of the angle θ between the incoming direction ω and outgoing ω' :

$$p(\mathbf{x}, \omega', \omega) = p(\mathbf{x}, \langle \omega' | \omega \rangle) = p(\mathbf{x}, \cos\theta)$$

Media with this type of phase functions are called **isotropic** because their response to incident illumination is invariant to rotations.

If scattering is equal towards all directions, i.e. if scattering is independent of incoming and outgoing direction the phase function is called **isotropic**. In this case we get

$$p(\cos(\theta)) = \frac{1}{4\pi}$$

The Henyey-Greenstein phase function¹⁾

The phase function developed by Henyey and Greenstein in 1941 controls the distribution of scattered light by a single parameter $g \in [-1, 1]$, called **assymetry parameter**:

$$p(x, \langle \omega' | \omega \rangle) = p(\cos(\theta)) = \frac{1}{4\pi} \frac{1 - g^2}{(1 + g^2 - 2g\cos(\theta))^{3/2}}$$

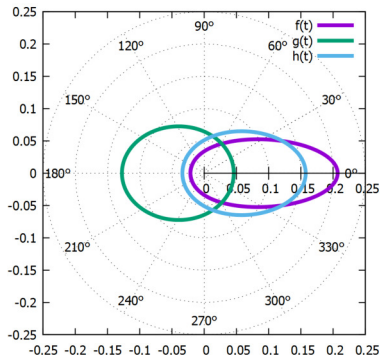


Figure: Examples of phase functions with different $g(f)=-0.67$, $g(g)=-0.5$ and $g(h)=0.35$.

The asymmetry parameter was carefully chosen as the cosine weighted average value of p :

$$g = \frac{1}{4\pi} \int_{\Omega} p(\mathbf{x}, \boldsymbol{\omega}', \boldsymbol{\omega}) \langle \boldsymbol{\omega} | \boldsymbol{\omega}' \rangle d\boldsymbol{\omega}'$$

Thus, an isotropic phase function gives $g = 0$, as expected.

More complex type of scattering can be modelled by a affine combination of several Henyey-Greenstein functions

$$p(\mathbf{x}, \langle \boldsymbol{\omega}' | \boldsymbol{\omega} \rangle) = \sum_{i=1}^N \frac{1}{4\pi} \alpha_i \frac{1 - g_i^2}{(1 + g_i^2 - 2g_i \cos(\theta))^{3/2}},$$

where $\sum_{i=1}^N \alpha_i = 1$.

An advantage of the Henyey-Greenstein phase function is that importance sampling of it can be done efficiently by distributing $\cos(\theta)$ according to

$$\cos(\theta) = \frac{1}{2g} \left(1 + g^2 - \frac{1 - g^2}{(1 - g^2 + 2g\xi)^2}, \right)$$

where ξ is a uniformly distributed random variable within $[0, 1]$.

To derive this result we need the accumulated distribution of $P(\cos(\theta))$:

$$\begin{aligned} P(\mu) &= \frac{1}{2} \int_{-1}^{\mu} \frac{1 - g^2}{(1 + g^2 - 2g\mu)^{3/2}} d\mu \\ &= \frac{1 - g^2}{2g} \left[(1 + g^2 - 2g\mu)^{-1/2} - (1 + g)^{-1} \right] \end{aligned}$$

Inverting the cumulative distribution we get the desired result.

The Henyey-Greenstein phase function

Blasi, le Sec and Schlick developed an efficient approximation to the Henyey-Greenstein function that is widely used in graphics due to its computational efficiency:

$$p_{schlick}(\cos(\theta)) = \frac{1}{4\pi} \frac{1 - k^2}{(1 - k\cos(\theta))^2},$$

where $k = 1.55g - 0.55g^3$.

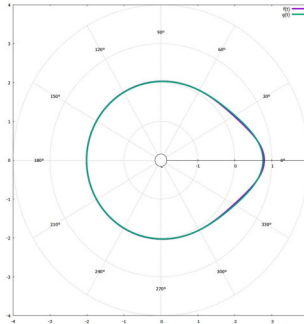


Figure: Schlick approximation g of Heneye-Greenstein function f with $g(f)=-0.6$.

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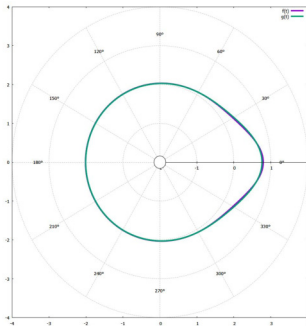
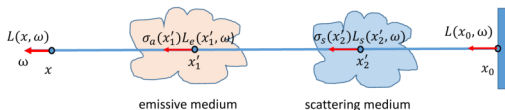


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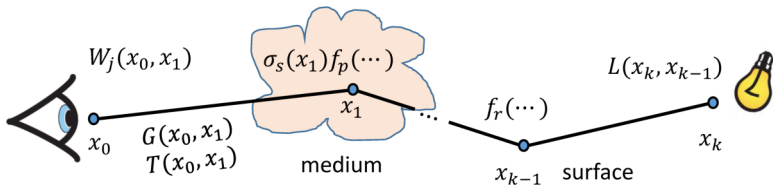
Reminder: Volume rendering equation



Using the transmittance $T(x_0, x) = e^{-\tau(x', x)}$ the volume rendering equation (VRE) reads

$$\begin{aligned} L(x, \omega) &= L(x_0, \omega)T(x_0, x) \\ &+ \int_0^s T(x', x) [\sigma_a(x')L_e(x', \omega) + \sigma_s(x')L_s(x', \omega)] dt \end{aligned}$$

Solving the VRE: path integral formulation



In the path integral formulation, the value of pixel j in the rendered image I can be described as a conceptually simple integral:

$$L_j = \int_{\mathcal{P}} f_j(\bar{x}) d\bar{x} \approx \underbrace{\frac{1}{N} \sum_{i=1}^N \frac{f_j(\bar{x}_i)}{p(\bar{x}_i)}}_{\text{pixel estimator}}$$

where $\bar{x} = (x_0, x_1, \dots, x_n) \in \mathcal{P}$ is a transport path of length k and $\mathcal{P}$ is the space of all such possible paths in the scene.

The measurement contribution function $f_j(\bar{x})$ gives the pixel response to the differential flux carried by the given path:

$$f_j(\bar{x}) = W(\mathbf{x}_0, \mathbf{x}_1) L_e(\mathbf{x}_k, \mathbf{x}_{k-1}) G(\mathbf{x}_0, \mathbf{x}_1) T(\mathbf{x}_0, \mathbf{x}_1) \cdot \prod_{i=1}^{k-1} f_s(\mathbf{x}_{x_i-1}, \mathbf{x}_i, \mathbf{x}_{i+1}) G(\mathbf{x}_i, \mathbf{x}_{i+1}) T(\mathbf{x}_i, \mathbf{x}_{i+1}),$$

where $W_j(\mathbf{x}_0, \mathbf{x}_1)$ is the response of pixel j , and

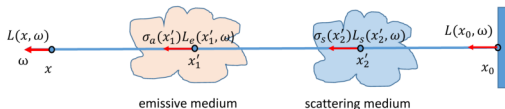
$$G(\mathbf{x}, \mathbf{x}') = \frac{D(\mathbf{x}, \mathbf{x}') D(\mathbf{x}', \mathbf{x})}{\|\mathbf{x} - \mathbf{x}'\|}, \text{ where}$$

$$D(\mathbf{x}, \mathbf{x}') = \begin{cases} |n(\mathbf{x}) \omega_{\mathbf{x} \rightarrow \mathbf{x}'}|, & \text{if } \mathbf{x} \text{ is on the surface,} \\ 1, & \text{if } \mathbf{x} \text{ is in a medium} \end{cases}$$

$$L_e(\mathbf{x}, \mathbf{x}') = \begin{cases} L_e(\mathbf{x}, \omega_{\mathbf{x} \rightarrow \mathbf{x}'}), & \text{if } \mathbf{x} \text{ is on a surface,} \\ \sigma_a L_e(\mathbf{x}, \omega_{\mathbf{x} \rightarrow \mathbf{x}'}), & \text{if } \mathbf{x} \text{ is in a medium} \end{cases}$$

$$f_s(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \begin{cases} f_r(\omega_{\mathbf{y} \rightarrow \mathbf{x}}, \mathbf{y}, \omega_{\mathbf{y} \rightarrow \mathbf{z}}), & \text{if } \mathbf{y} \text{ is on a surface,} \\ \sigma_a f_p(\omega_{\mathbf{y} \rightarrow \mathbf{x}}, \mathbf{y}, \omega_{\mathbf{y} \rightarrow \mathbf{z}}), & \text{if } \mathbf{y} \text{ is in a medium.} \end{cases}$$

Reminder: Volume rendering equation



A Monte Carlo estimate of the VRE is given by

$$L(\mathbf{x}, \omega) = \frac{T(\mathbf{x}, \mathbf{x}_0)}{p(\mathbf{x}_0)} L(\mathbf{x}_0, \omega) + \frac{T(\mathbf{x}, \mathbf{x}')}{p(\mathbf{x}')} [\sigma_a(\mathbf{x}') L_e(\mathbf{x}', \omega) + \sigma_s(\mathbf{x}') L_s(\mathbf{x}', \omega)],$$

where $p(\mathbf{x}')$ is the PDF of sampling $\mathbf{x}'$ and where the nearest-surface radiance can be optionally evaluated with probability $p(\mathbf{x}_0)$.

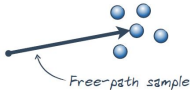
This estimator requires two main routines

- ▶ sampling distances along the ray,
- ▶ evaluating the transmittance T between two given points.



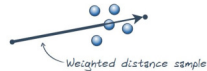
Analog methods

- ▶ Adhere to physical process
- ▶ Produce free-path samples
- ▶ Energy of particles unchanged



Non-Analog methods

- ▶ Deviate from physical process
- ▶ Produce arbitrary distance samples
- ▶ Particles (photons) are weighted





How to sample the flight distance t to the next interaction?

The transmittance $T(t)$ gives, in expectation, the fraction of photons, that travel beyond the distance t without colliding with the medium.

Denoting X a random **free-path** variable, we demand

$$P(X > s) = \tau(s) = \tau(\mathbf{x}_0, \mathbf{x} = \mathbf{x}_0 + s\boldsymbol{\omega}) = \int_0^s \sigma_t(\mathbf{x}_0 + t\boldsymbol{\omega}, \boldsymbol{\omega}) dt$$

Let $F(s)$ be the cumulative distribution function (CDF) of X , i.e.

$$F(s) := P(X \leq s)$$

then

$$F(s) = 1 - \tau(s).$$



Approaches for finding t :

1. Analytic closed-form CDF^{-1}
2. Semi-Analytic (regular tracking)
3. Approximate (ray marching)

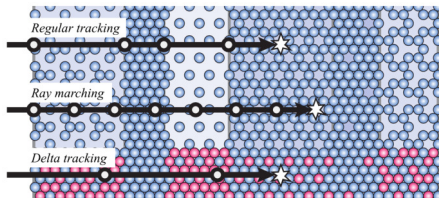


Figure: Illustration of free-path sampling in a medium composed of several homogeneous regions; blue and pink particles represent real and fictitious matter, respectively. Regular tracking finds all boundary crossings and solves for the position analytically. Ray marching steps with a constant stride until it accumulates the sampled optical thickness. Delta tracking first homogenizes the volume using fictitious matter and then compensates for that by explicitly handling the resulting null collisions with that matter.²

²J. Novák et al. "Monte Carlo Methods for Volumetric Light Transport Simulation". en. In: *Computer Graphics Forum* 37.2 (May 2018), pp. 551–576. URL: <http://doi.wiley.com/10.1111/cgf.13383> (visited on 06/18/2021).



The probability density function (**PDF**) $p(t)$ of F is given by

$$p(t) = \frac{dF(t)}{dt} = \frac{d}{dt}(1 - e^{-\tau(t)}) = \sigma_t(t)e^{-\tau(t)}$$

Knowing the cumulative distribution function (**CDF**)

$$F(t) = 1 - e^{-\tau(t)}$$

of the free-path variable X we have to compute the inverted cumulative distribution function (**CDF**⁻¹) to generate samples:

$$\xi = 1 - e^{-\tau(t)} \Leftrightarrow \tau(t) = -\ln(1 - \xi)$$

$$\int_0^t \sigma_t(s)ds = -\ln(1 - \xi)$$

Unfortunately, in the general case the inversion method only yields this implicit equation and there is no straightforward Monte Carlo estimator available to compute t .



Some simple volumes permit closed-form solutions

Example: **homogeneous** medium ($\sigma_t(x) = \sigma_t$)

$$\int_0^t \sigma_t(s) ds = \sigma_t t \quad \Leftrightarrow \quad t(\xi) = -\frac{\ln(1 - \xi)}{\sigma_t}, \xi \in [0, 1].$$

$$\int_0^t \sigma_t(s) ds = t\sigma_t \Leftrightarrow F^{-1}(\xi) = -\frac{\ln(1 - \xi)}{\sigma_t}$$

a.k.a. Woodcock tracking, pseudo scattering, hole tracking, null-collision method,...

Add FICTITIOUS MATTER to homogenize heterogeneous extinction

- ▶ albedo $\alpha(\mathbf{x}) = 1$
- ▶ phase function $p(\mathbf{x}, \omega, \omega') = \delta(\omega - \omega')$

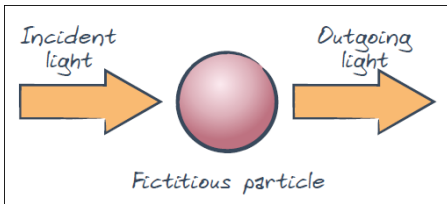


Figure: Physical interpretation: presence of fictitious matter does not impact light transport³

³J. Novák et al. "Monte Carlo Methods for Volumetric Light Transport Simulation". en. In: *Computer Graphics Forum* 37.2 (May 2018), pp. 551–576. URL: <http://doi.wiley.com/10.1111/cgf.13383> (visited on 06/18/2021).

Change of radiance due to null collisions is formalized by

$$\underbrace{-\sigma_n(\mathbf{x})L(\mathbf{x}, \boldsymbol{\omega})}_{\text{Losses}} + \underbrace{\sigma_n(\mathbf{x}) \int_{\Omega} \delta(\boldsymbol{\omega} - \boldsymbol{\omega}') L(\mathbf{x}, \boldsymbol{\omega}') d\boldsymbol{\omega}'}_{\text{Gains ("inscattering")}} = 0$$

Adding the left hand side to the **Radiative Transfer Equation** leads to:

$$\begin{aligned} (\boldsymbol{\omega} \cdot \nabla) L(\mathbf{x}, \boldsymbol{\omega}) &= \sigma_a(\mathbf{x}) L_e(\mathbf{x}, \boldsymbol{\omega}) - (\sigma_t + \sigma_n)(\mathbf{x}) L(\mathbf{x}, \boldsymbol{\omega}) \\ &+ (\sigma_s + \sigma_n)(\mathbf{x}) \int_{\Omega} (p(\mathbf{x}, \boldsymbol{\omega}', \boldsymbol{\omega}) + \delta(\boldsymbol{\omega} - \boldsymbol{\omega}')) L_i(\mathbf{x}, \boldsymbol{\omega}') d\boldsymbol{\omega}' \end{aligned}$$

Integrating this equation leads to the **RTE with null collisions**

$$\begin{aligned} L(\mathbf{x}, \boldsymbol{\omega}) &= L(\mathbf{x}_0, \boldsymbol{\omega}) T(\mathbf{x}_0, \mathbf{x}) \\ &+ \int_0^s T_{\bar{\sigma}}(\mathbf{x}', \mathbf{x}) [\sigma_a(\mathbf{x}') L_e(\mathbf{x}', \boldsymbol{\omega}) + \sigma_s(\mathbf{x}') L_i(\mathbf{x}', \boldsymbol{\omega}) + \sigma_n(\mathbf{x}') L(\mathbf{x}', \boldsymbol{\omega})] dt, \end{aligned}$$

where $\bar{\sigma} = \sigma_t + \sigma_n$ is the total collision coefficient, which is spatially invariant.

Theorem (Coleman, 1963⁴)

Let $\bar{\sigma} = \sigma_t(x) + \sigma_n(x) \geq \sigma_t(x)$ for all $x \in \mathcal{V}$, let $t_1, t_2, \dots$, be independent random distances sampled according to the closed form solution with parameter $\bar{\sigma}$ and let $\xi_1, \xi_2, \dots \in [0, 1]$ be independent uniformly distributed numbers. Then, the first distance $T_{i_0} = \sum_{i=1}^{i_0} t_i$, satisfying $\xi_{i_0} \leq \frac{\sigma_t(x_0 + T_{i_0}\omega)}{\bar{\sigma}}$ is distributed according to

$$F(t) = \int_0^t \sigma_t(s) e^{-\int_0^s \sigma_t(u) du} ds.$$

⁴W. Coleman. "Mathematical verification of a certain Monte Carlo sampling technique and applications of the technique to radiation transport problems". In: *Nuclear science and engineering* 32.1 (1968), pp. 76–81.

This lead to the following algorithm⁵ to sampling the distance:

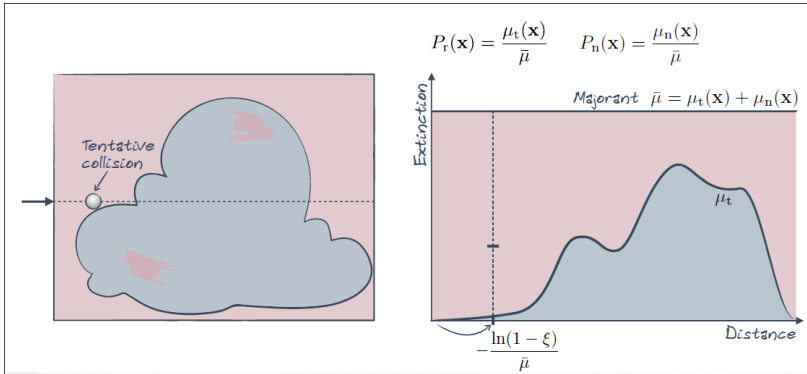
```
float sampleDistance(Point  $x_0$ , Direction  $\omega$ )
{
    //sample with the maximum extinction  $\sigma_t$ 
    float  $t = -\log(\text{rand}()) / \sigma_t$ ;

    while ( $\frac{\sigma_t(x_0+t\omega)}{\sigma_t} < \text{rand}()$ )
         $t -= \log(\text{rand}()) / \sigma_t$ ;

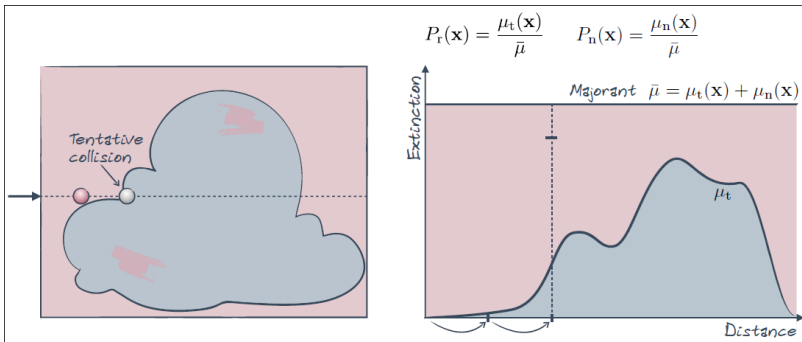
    return  $t$ ;
}
```

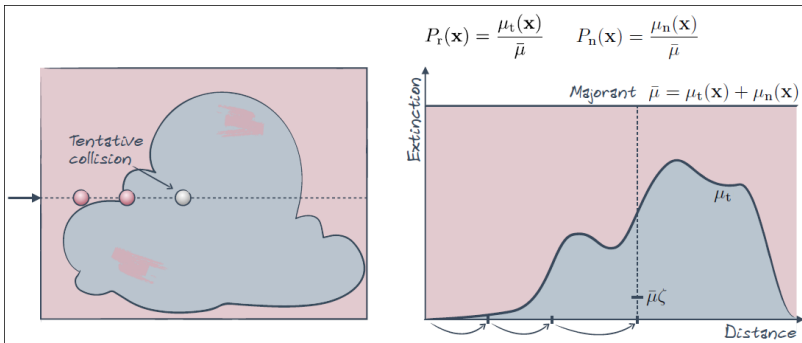
⁵M. Raab et al. “Unbiased global illumination with participating media”. In: *Monte Carlo and Quasi-Monte Carlo Methods 2006*. Springer, 2008, pp. 591–605.

Illustration of delta tracking⁶ (note that Novak use μ instead of σ to denote the extinction coefficient)

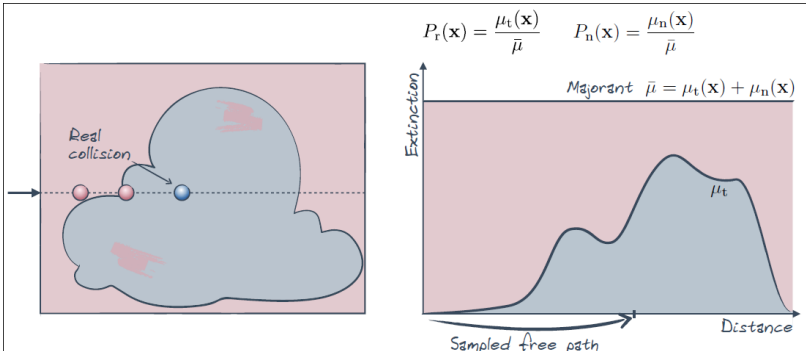


⁶J. Novák et al. "Monte Carlo Methods for Volumetric Light Transport Simulation". en. In: *Computer Graphics Forum* 37.2 (May 2018), pp. 551–576. URL: <http://doi.wiley.com/10.1111/cgf.13383> (visited on 06/18/2021).





Null-collision methods: Delta tracking





Until now, we only sampled the transmittance but we need also be able to compute it to properly attenuate a ray traveling through the medium (often needed for NEE).

In the general case, no closed-form solution exists, so use Monte-Carlo Integration.

Reminder: Transmittance $T(a, b)$ is the probability of reaching b without scattering, therefore we can formulate two estimators:

Delta-Tracking:

1. Do delta tracking until a scattering event x_{k+1} was sampled.
- 2.

$$\langle T(a, b) \rangle = \begin{cases} 1 & \text{if } b \leq x_{k+1}, \\ 0 & \text{otherwise} \end{cases}$$

Ratio-Tracking:

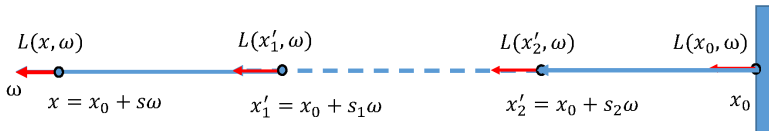
1. Sample null-scattering events like in Delta-Tracking but don't reject sample before b , deterministically trace until b is reached
- 2.

$$\langle T(a, b) \rangle = \prod_{i=1}^k \frac{\mu_n(x_i)}{\bar{\mu}(x_i)}$$

where x_k is the last collision before reaching b : $x_k < b \leq x_{k+1}$.



In 2019 Georgiev et al.⁷ came up with an integral formulation of the transmittance.



We first observe that

$$T(\mathbf{x}'_1, \mathbf{x}'_2) = \frac{L(\mathbf{x}'_1)}{L(\mathbf{x}'_2)}$$

⁷I. Georgiev et al. "Integral formulations of volumetric transmittance". In: *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia)* 38.6 (Nov. 2019).



In addition we get

$$\begin{aligned}(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}, \boldsymbol{\omega}) &= -\sigma_t(\boldsymbol{x})L(\boldsymbol{x}, \boldsymbol{\omega}) \\ \int_{s_1}^{s_2} -(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(\boldsymbol{x}(s), \boldsymbol{\omega})ds &= \int_{s_1}^{s_2} -\sigma_t(\boldsymbol{x}(s))L(\boldsymbol{x}(s), \boldsymbol{\omega})ds \\ L(\boldsymbol{x}'_1) - L(\boldsymbol{x}'_2) &= \int_{s_1}^{s_2} -\sigma_t(\boldsymbol{x})L(\boldsymbol{x}(s), \boldsymbol{\omega})ds\end{aligned}$$

Deviding by $L(\boldsymbol{x}'_2)$ leads to the

Integral formulation

$$T(\boldsymbol{x}'_1, \boldsymbol{x}'_2) = 1 + \int_{s_1}^{s_2} -\sigma_t(\boldsymbol{x})T(\boldsymbol{x}(s), \boldsymbol{x}'_2)ds$$



For an infinitesimal beam entering a homogeneous medium, the incoming radiance will decrease exponentially with distance s .

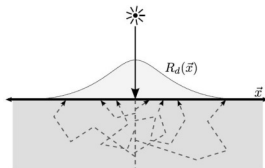


Figure: If a focused, infinitesimally thin pencil beam in the origin strikes the surface of a half-infinite slab at normal incident the photons originating in the beam travel along the refracted ray until they are scattered by the medium, leading to a series of scattering events until they are ultimately absorbed or escape through the surface.

This is referred to as the *reduced intensity*:

$$L_r(x_i + s, \omega) = e^{-\sigma_t s} \cdot L_i(x_i, \omega_i)$$

The first-order scattering of the reduced intensity, L_{ri} , may be treated as a volumetric source:

$$Q(x, \omega) = \sigma_s \int_{\Omega} p(\omega', \omega) L_{ri}(x, \omega') d\omega'.$$



Although R_d can be computed numerically using Monte Carlo integration the computational costs are very high. This motivates finding closed-form approximations for R_d :

In the case of homogeneous, optically thick, highly scattering materials (i.e., those with large albedos) the transfer equation can be approximated by the simpler

Diffusion Equation.

With an isotropic phase function $p = 1/4\pi$, the transfer equation

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(x, \boldsymbol{\omega}) = \sigma_a(x)L_e(x, \boldsymbol{\omega}) - \sigma_t(x)L(x, \boldsymbol{\omega}) + \sigma_s(x) \int_{\Omega} p(x, \boldsymbol{\omega}', \boldsymbol{\omega}) L_i(x, \boldsymbol{\omega}') d\boldsymbol{\omega}'$$

simplifies to

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(x, \boldsymbol{\omega}) = \sigma_a(x)L_e(x, \boldsymbol{\omega}) - \sigma_t(x)L(x, \boldsymbol{\omega}) + \frac{\sigma_s(x)}{4\pi} \int_{\Omega} L(x, \boldsymbol{\omega}') d\boldsymbol{\omega}'$$

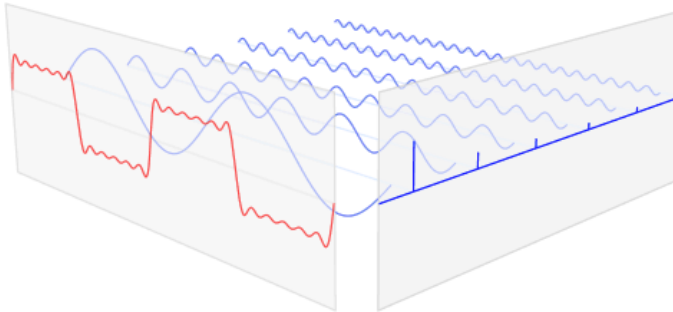


Spherical Harmonics





Reminder: Consider the Fourier Transform:



Spatial domain

$$f(t) = \int_{\mathbb{R}} \hat{f}(\omega) e^{2\pi i \omega t} d\omega$$

Frequency domain

$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-2\pi i \omega t} dt$$



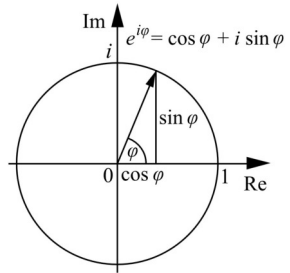
The Fourier Transform was derived as the projection of f to sine and cosine functions:

Lemma (Euler's formula)

The equation

$$e^{ix} = \cos(x) + i \sin(x)$$

is called Euler's formula.



So we can write the complex sum of a sine and cosine as an exponential of a complex number. Furthermore, we can use the fact that the functions

$$e_{\omega}: t \mapsto e^{2\pi i \omega t}$$

span an **orthonormal basis** over the so called **frequency domain**. Since this basis is orthonormal, we can compute the component of f to each basis function using the dot product.



Definition (Harmonic function)

A function f is called *harmonic* if it fulfills the Laplace-Equation:

$$\Delta f = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} = 0$$

To formulate a theorem about basis functions, we need the notion of an n -sphere:

$$\mathcal{S}^n := \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \|\mathbf{x}\| = r\}$$

Theorem (Harmonic functions on $\mathcal{S}^1$)

The Fourier Basis Functions

$$e_\omega: t \mapsto e^{2\pi i \omega t}$$

are homogeneous solutions of the Laplace-Equation in polar coordinates and form an orthonormal basis on $\mathcal{S}^1$.



Objective: Extend the concept of frequency domain to the sphere $\mathcal{S}^2$.

Step 1: Extend the polar coordinates to spherical coordinates:

$$s_1: [0, 2\pi] \rightarrow \mathcal{S}^1$$

$$s_2: [0, 2\pi] \times [0, \pi] \rightarrow \mathcal{S}^2$$

$$\phi \mapsto \begin{pmatrix} \cos(\phi) \\ \sin(\phi) \end{pmatrix}$$

$$(\phi, \theta) \mapsto \begin{pmatrix} \cos(\phi) \sin(\theta) \\ \sin(\phi) \sin(\theta) \\ \cos(\theta) \end{pmatrix}$$

Step 2: Reformulate the Fourier Basis Functions to fit the new domain:

$$e_\omega: [0, \infty] \rightarrow \mathbb{C}$$

$$e_m: [0, 2\pi] \rightarrow \mathbb{C}$$

$$t \mapsto e^{2\pi i \omega t}$$

$$\phi \mapsto \frac{1}{\sqrt{2\pi}} e^{im\phi}$$

$$\omega \equiv \text{ordinary frequency} \left[\frac{1}{\text{s}} \right]$$

$$m \equiv \text{angular frequency} \left[\frac{\text{rad}}{\text{s}} \right]$$

Step 3: Define a dot product on the sphere:

$$\langle f | g \rangle_{\mathcal{H}} := \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} f(\phi, \theta) \overline{g(\phi, \theta)} \sin(\theta) d\theta d\phi$$



Step 4: Define Basis Functions on $\mathcal{S}^2$:

Theorem (Harmonic functions on $\mathcal{S}^2$)

The Spherical Harmonics

$$Y_l^m : [0, 2\pi] \times [0, \pi] \rightarrow \mathbb{C}$$

$$(\phi, \theta) \mapsto \frac{1}{\sqrt{2\pi}} N_l^m P_l^m(\cos(\theta)) e^{im\phi}$$

are homogeneous solutions of the Laplace-Equation in spherical coordinates and form an orthonormal basis on $\mathcal{S}^2$.

The Spherical Harmonics can be seen as the tensor product of the Fourier Basis Functions $\frac{1}{\sqrt{2\pi}} e^{im\phi}$ and the Associated Legendre Polynomials $P_l^m(\cos(\theta))$ normalized by the constant

$$N_l^m = \sqrt{\frac{2l+1}{2} \frac{(l-m)!}{(l+m)!}}$$

Note: Here, the frequency bands are the integer numbers l !



The Associated Legendre Polynomials P_l^m are solutions to the General Legendre Equation and defined recursively:

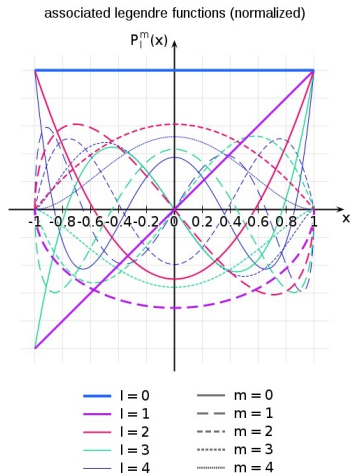
$$P_m^m(x) = (-1)^m (2m-1)!! (1-x^2)^{m/2}$$

$$P_{m+1}^m(x) = x(2m+1)P_m^m$$

$$P_l^m(x) = x \frac{2l-1}{l-m} P_{l-1}^m - \frac{l+m-1}{l-m} P_{l-2}^m$$

with

$$(2m-1)!! := \prod_{i=1}^m (2i-1)$$





They form an orthogonal basis over the interval $[-1, 1]$:

$$\int_{-1}^1 P_k^m(x) P_l^m(x) dx = \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \delta_{kl} = \frac{\delta_{kl}}{(N_l^m)^2}$$

Therefore, the Spherical Harmonics as defined above are also orthonormal. Normally, we only need a real valued basis:

$$\begin{aligned} y_l^m(\phi, \theta) &= \begin{cases} \sqrt{2} \operatorname{Re}(Y_l^m(\phi, \theta)) & \text{if } m > 0 \\ \sqrt{2} \operatorname{Im}(Y_l^m(\phi, \theta)) & \text{if } m < 0 \\ Y_l^0(\phi, \theta) & \text{if } m = 0 \end{cases} \\ &= \begin{cases} \sqrt{2} K_l^m \cos(m\phi) P_l^m(\cos(\theta)) & \text{if } m > 0 \\ \sqrt{2} K_l^{|m|} \sin(|m|\phi) P_l^{|m|}(\cos(\theta)) & \text{if } m < 0 \\ K_l^0 P_l^0(\cos(\theta)) & \text{if } m = 0 \end{cases} \end{aligned}$$

with

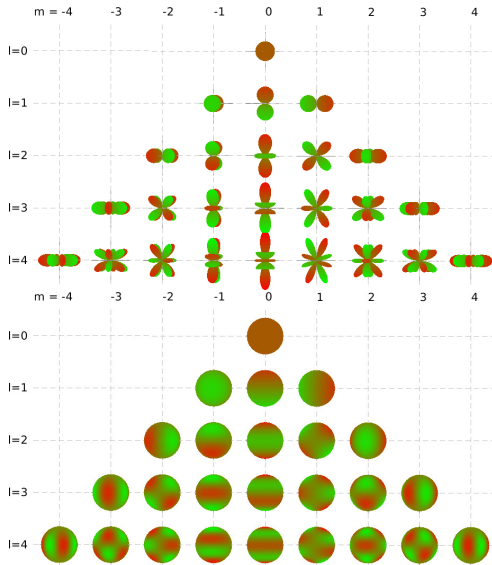
$$K_l^m = \frac{1}{\sqrt{2\pi}} N_l^m = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}}$$



Additionally, we can simplify the indexing:

$$y_l^m(\phi, \theta) = y_i(\phi, \theta),$$
$$i = l(l + 1) + m$$

Visualization:



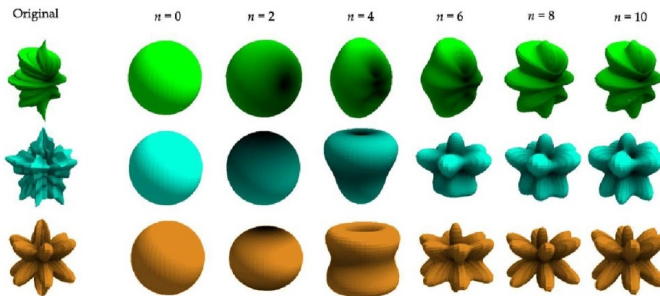


The projection to this (real-valued) basis is done by using the dot product:

$$\hat{f}_i = \langle f | y_i \rangle_{\mathcal{H}} := \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} f(\phi, \theta) y_i(\phi, \theta) \sin(\theta) d\theta d\phi.$$

The function can be reconstructed using the first n bands by

$$f(\phi, \theta) \approx \sum_{l=0}^{n-1} \sum_{m=-l}^l \hat{f}_l^m y_l^m(\phi, \theta) = \sum_{i=0}^{n^2} \hat{f}_i y_i(\phi, \theta).$$





Now, we will derive a very important property:

Definition (Unitary Transformation)

A transformation $U : H_1 \rightarrow H_2$ between two vector spaces H_1 and H_2 , each having a dot product, is called *unitary* if

$$\langle Ux | Uy \rangle_{H_2} = \langle x | y \rangle_{H_1}$$

We will prove this for the Spherical Harmonics Transform:

Theorem (Spherical Harmonics Transform)

Let $\hat{a}, \hat{b}$ be the transformations of the functions a, b to the spherical frequency space using the (real-valued) Spherical Harmonics Transform. Then this transformation is unitary:

$$\langle \hat{a} | \hat{b} \rangle = \langle a | b \rangle_{\mathcal{H}}$$



Proof.

The transformations $\hat{a}$ and $\hat{b}$ are defined as

$$\hat{a}_i = \langle a | y_i \rangle_{\mathcal{H}} \quad , \quad \hat{b}_i = \langle b | y_i \rangle_{\mathcal{H}}$$

So we can rewrite a and b using these coefficients:

$$a(\phi, \theta) = \sum_{i=0}^{\infty} \hat{a}_i y_i(\phi, \theta) \quad , \quad b(\phi, \theta) = \sum_{i=0}^{\infty} \hat{b}_i y_i(\phi, \theta)$$

Now, we can conclude:

$$\begin{aligned} \langle a | b \rangle_{\mathcal{H}} &= \left\langle \sum_{i=0}^{\infty} \hat{a}_i y_i \left| \sum_{j=0}^{\infty} \hat{b}_j y_j \right. \right\rangle_{\mathcal{H}} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \hat{a}_i \hat{b}_j \langle y_i | y_j \rangle_{\mathcal{H}} \\ &= \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \hat{a}_i \hat{b}_j \delta_{ij} = \sum_{i=0}^{\infty} \hat{a}_i \hat{b}_i = \langle \hat{a} | \hat{b} \rangle \end{aligned}$$

□



Writing the radiance $L(x, \omega) = L(\omega)$ at position x along a ray in terms of the zeros and first order spherical harmonic basis functions $Y_{l,m}$, $l = 0, 1$, $m = -l, l$ and omitting higher order terms we get

$$\begin{aligned} L(x, \omega) &\approx \frac{1}{4\pi} \int_{\Omega} L(\omega) d\omega + \frac{3}{4\pi} \omega \cdot \int_{\Omega} \omega L(\omega) d\omega \\ &= \frac{1}{4\pi} \int_{\Omega} L(\omega) d\omega + \frac{3}{4\pi} \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} \cdot \int_0^{2\pi} \int_0^{\pi} \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} L(\theta, \varphi)\sin(\theta) d\theta d\varphi \\ &= \frac{1}{4\pi} \Phi(x) + \frac{3}{4\pi} \omega \cdot E(x). \end{aligned}$$



The scalar term $\Phi(x)$ in the approximation

$$L(x, \omega) \approx \frac{1}{4\pi} \Phi(x) + \frac{3}{4\pi} \omega \cdot E(x)$$

is called **fluence** and represents the isotropic flux, defined as the zero order moment of L :

$$\Phi(x) = \mu_0(L(x, \cdot)) = \int_{\Omega} L(\omega) d\omega$$

The vector Term $E(x)$, called **vector irradiance** or **vector flux**, represents the anisotropic flux and can be thought of as the net photon current at a point x in the material, defined as the first order moment of L

$$\begin{aligned} \mu_1(L(x, \cdot)) &= \int_{\Omega} \omega L(\omega) d\omega \\ &= \int_0^{2\pi} \int_0^{\pi} \begin{pmatrix} \cos(\varphi)\sin(\theta) \\ \sin(\varphi)\sin(\theta) \\ \cos(\theta) \end{pmatrix} L(\theta, \varphi) \sin(\theta) d\theta d\varphi \end{aligned}$$

is the first moment of $L(x, \omega)$.



To derive this result we use the definition of the SH basis functions

$$Y_{0,0}(\theta, \varphi) = \sqrt{\frac{1}{4\pi}}$$

$$Y_{1,-1}(\theta, \varphi) = \sqrt{\frac{3}{8\pi}} \sin(\theta) e^{-i\varphi}, Y_{1,0}(\theta, \varphi) = \sqrt{\frac{3}{4\pi}} \cos(\theta), Y_{1,1}(\theta, \varphi) = -\sqrt{\frac{3}{8\pi}} \sin(\theta) e^{i\varphi}$$

to write

$$\begin{aligned} L(\theta, \varphi) &= \sum_{l=0}^1 \sum_{m=-1}^1 c_{lm} Y_{lm}(\theta, \varphi) \\ &= c_{0,0} \sqrt{\frac{1}{4\pi}} + c_{1,-1} \sqrt{\frac{3}{8\pi}} \sin(\theta) e^{-i\varphi} + c_{1,0} \sqrt{\frac{3}{4\pi}} \cos(\theta) - c_{1,1} \sqrt{\frac{3}{8\pi}} \sin(\theta) e^{i\varphi} \end{aligned}$$



The coefficients $c_{l,m}$ are given by

$$c_{l,m} = \int_{\Omega} Y_{l,m}^*(\omega) L(\omega) d\omega = \int_{\Omega} Y_{l,m}^*(\theta, \varphi) L(\theta, \varphi) \sin(\theta) d\theta d\varphi$$

Using this formula we get

$$c_{0,0} Y_{0,0}(\theta, \varphi) = \frac{1}{4\pi} \int_{\Omega} L(\omega) d\omega$$

$$c_{1,0} Y_{1,0}(\theta, \varphi) = \frac{3}{4\pi} \frac{1}{2} \int_0^{2\pi} \int_0^{\pi} \cos(\theta) L(\theta, \varphi) d\theta d\varphi \cdot \cos(\theta)$$

$$c_{1,-1} Y_{1,-1}(\theta, \varphi) = \frac{3}{4\pi} \frac{1}{2} \int_0^{2\pi} \int_0^{\pi} \sin(\theta) (\cos(\varphi) + i \sin(\varphi)) L(\theta, \varphi) \sin(\theta) d\theta d\varphi \cdot \sin(\theta) e^{-i\varphi}$$

$$c_{1,1} Y_{1,1}(\theta, \varphi) = \frac{3}{4\pi} \frac{1}{2} \int_0^{2\pi} \int_0^{\pi} \sin(\theta) (\cos(\varphi) - i \sin(\varphi)) L(\theta, \varphi) \sin(\theta) d\theta d\varphi \cdot \sin(\theta) e^{i\varphi}$$



Summing up the terms in the $l = 1$ band observing the Euler identities
 $\cos(\varphi) = \frac{1}{2}(e^{-i\varphi} + e^{i\varphi})$ and $-i\sin(\varphi) = \frac{1}{2}(e^{-i\varphi} - e^{i\varphi})$
 leads to

$$\begin{aligned}
 & \frac{3}{4\pi} \left(\int_0^{2\pi} \int_0^\pi L(\theta, \varphi) \underbrace{\cos(\varphi)\sin(\theta)}_{\omega_x} \sin(\theta) d\theta d\varphi \cdot \underbrace{\cos(\varphi)\sin(\theta)}_{\omega_x} \right. \\
 & + \int_0^{2\pi} \int_0^\pi L(\theta, \varphi) \underbrace{\sin(\varphi)\sin(\theta)}_{\omega_y} \sin(\theta) d\theta d\varphi \cdot \underbrace{\sin(\varphi)\sin(\theta)}_{\omega_y} \\
 & \left. + \int_0^{2\pi} \int_0^\pi L(\theta, \varphi) \underbrace{\cos(\theta)}_{\omega_z} \sin(\theta) d\theta d\varphi \cdot \underbrace{\cos(\theta)}_{\omega_z} \right)
 \end{aligned}$$

Adding the term in the $l = 0$ band gives the desired result.



Writing both sides of the simplified transfer equation

$$(\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(x, \boldsymbol{\omega}) = \sigma_a(x)L_e(x, \boldsymbol{\omega}) - \sigma_t(x)L(x, \boldsymbol{\omega}) + \frac{\sigma_s(x)}{4\pi} \int_{\Omega} L(x, \boldsymbol{\omega}') d\boldsymbol{\omega}'$$

in the SH-basis requires the coefficients $c_{l,m}$ to be the same, i.e. for $i = 1, 2$

$$\mu_i(\boldsymbol{\omega} \cdot \boldsymbol{\nabla}L(x, \boldsymbol{\omega})) = \mu_i \left(\sigma_a(x)L_e(x, \boldsymbol{\omega}) - \sigma_t(x)L(x, \boldsymbol{\omega}) + \frac{\sigma_s(x)}{4\pi} \int_{\Omega} L(x, \boldsymbol{\omega}') d\boldsymbol{\omega}', \right).$$

On the left hand side this results for the zeroth moment in

$$\begin{aligned} \int_{\Omega} (\boldsymbol{\omega} \cdot \boldsymbol{\nabla})L(x, \boldsymbol{\omega}) &= \int_{\Omega} \left(\omega_x \frac{\partial L(x, \boldsymbol{\omega})}{\partial x} + \omega_y \frac{\partial L(x, \boldsymbol{\omega})}{\partial y} + \omega_z \frac{\partial L(x, \boldsymbol{\omega})}{\partial z} \right) \\ &= \operatorname{div} E(x). \end{aligned}$$



Calculating the right hand side of the equation we get

$$\begin{aligned}\operatorname{div} E(x) &= \sigma_a(x) \int_{\Omega} L_e(x, \omega) d\omega - \sigma_t(x) \int_{\Omega} L(x, \omega) d\omega \\ &\quad + \frac{\sigma_s(x)}{4\pi} \int_{\Omega} L(x, \omega') d\omega' \int_{\Omega} d\omega \\ &= \sigma_a(x) Q_0(x) - \sigma_t(x) \Phi(x) + \sigma_s(x) \Phi(x) \\ &= \sigma_a(x) Q_0(x) - \sigma_a \Phi(x),\end{aligned}$$

where $Q_0 = \mu_0(L_e)$ is the zeroth order approximation of the emitted light.

This equation states that the divergence of the irradiance vector field E is negative in the presence of absorption and positive when light is being added by Q_0 .

Similar calculations for the first order moments result in

$$\frac{1}{3} \nabla \Phi(x) = -\sigma_t E(x) + Q_1(x)$$

where $Q_1(x) = \mu_1(L_e(x))$ is the first order moment of the emitted light. If the light source in the medium emit light uniformly in all direction $Q_1(x) = 0$.



Solving the first order moment equation for $E(x)$ and substituting it into the zeroth order approximation yields the

Diffusion Equation

which only involves the fluence rate $\Phi(x)$:

$$\frac{1}{3\sigma_t} \text{div} \nabla \Phi(x) = \sigma_a(x) \Phi(x) - Q_0(x) + \frac{1}{3\sigma_t} \nabla \cdot Q_1(x).$$

Assuming $Q_1(x) = 0$, the diffusion equation can be written more compactly as

$$D \nabla^2 \Phi(x) - \sigma_a(x) \Phi(x) = -Q_0(x)$$

where $D = \frac{1}{3\sigma_t}$ is the **diffusion coefficient** and ∇^2 is the Laplace operator.



We can formulate the diffuse reflectance profile R_d in terms of the fluence Φ . R_d is defined as the radiant exitance divided by the incident flux. The radiant exitance is $\mathbf{n}(x) \cdot \mathbf{E}(x)$ and, using Fick's first law $\mathbf{E}(x) = -D\nabla\Phi(x)$, can be expressed in terms of the fluence⁸:

$$R_d(\|x_0 - x_i\|) = -D \frac{(\mathbf{n}(x) \cdot \nabla\Phi(x))(x_0)}{d\Phi_i(x_i)}$$

Here, D is the diffusion constant:

$$D = \frac{1}{3\sigma'_t}$$

This equation makes use of the reduced extinction coefficient, σ'_t . For isotropic scattering ($g = 0$), the reduced extinction coefficient is equal to the extinction coefficient σ_t . For anisotropic scattering ($g \neq 0$), the reduced extinction coefficient is:

$$\begin{aligned}\sigma'_t &= \sigma'_s + \sigma_a \\ \sigma'_s &= \sigma_s(1 - g)\end{aligned}$$

⁸H. W. Jensen et al. "A practical model for subsurface light transport". en. In: *Proceedings of the 28th annual conference on Computer graphics and interactive techniques - SIGGRAPH '01*. Not Known: ACM Press, 2001, pp. 511–518. URL: <http://portal.acm.org/citation.cfm?doid=383259.383319> (visited on 07/03/2019).



For an infinite medium with a unit power, isotropic point light source at the origin, i.e.

$$L_e(x, \omega) = \frac{1}{4\pi} \delta(x)$$

with corresponding moments

$$Q_i(x) = \begin{cases} \delta(x), & i = 0 \\ 0, & i = 1 \end{cases} \quad (1)$$

the diffusion equation has a simple solution⁹

$$\Phi(r(x)) = \frac{1}{4\pi D} \frac{e^{-\sigma_{tr}r(x)}}{r(x)},$$

where $r(x)$ is the distance from the light source and $\sigma_{tr} = \sqrt{\sigma_a/D}$ is the **effective transport coefficient**; it occurs in an exponential falloff term that accounts for absorption in the medium. Note that $\sigma_{tr} \neq \sigma_t$: instead, this modified attenuation coefficient additionally depends on the albedo to model the effect of multiple scattering ~~inside the medium, hence the term effective.~~

⁹H. W. Jensen et al. "A practical model for subsurface light transport". en. In: *Proceedings of the 28th annual conference on Computer graphics and interactive techniques - SIGGRAPH '01*. Not Known: ACM Press, 2001, pp. 511–518. URL: <http://portal.acm.org/citation.cfm?doid=383259.383319> (visited on 07/03/2019).



To verify this solution we observe that the Laplace operator of a function f that depends on the radius in spherical coordinates is given by

$$\nabla^2 f = r^{-2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} f(r) \right).$$

Taking the inner derivative of $\Phi(r(x))$ and multiplying by r^2 yields

$$r^2 \frac{\partial}{\partial r} \Phi(r(x)) = -r\Phi(r(x))(1 + \sigma_{tr}r).$$

Taking the outer derivative, multiplying by r^{-2} , and simplifying gives

$$r^{-2} \frac{\partial}{\partial r} (-r\Phi(r(x))(1 + \sigma_{tr})) = \sigma_{tr}^2 \Phi(r(x)) = \frac{\sigma_a}{D} \Phi(r(x))$$

which results exactly in

$$D\nabla^2 \Phi - \sigma_a \Phi = 0.$$



The monopole equation results in bad approximations when the assumptions made in the model are violated:

- ▶ Absorption prevents the radiance from reaching an isotropic equilibrium distribution,
- ▶ Close to the source, the true radiance function is dominated by a (highly non-isotropic) spherical Dirac delta function.

In 2011 Eugene d'Eon and Geoffrey Irving¹⁰ introduced a diffusion model developed by Grosjean in 1956¹¹ in the field of neutron transport into computer graphics in 2011:

$$\Phi_G(r(x)) = \frac{e^{-\sigma'_t r}}{4\pi r^2} + \Phi_M(\tilde{r}(x))$$

where Φ_M denotes the monopole solution.

The first term in this solution separates out an attenuated source term modeled using standard radiative transport—this effectively removes the portion that cannot easily be handled by diffusion. The remainder is a scaled diffusive term that accounts for light which has been scattered at least once, where the reduced albedo ρ' accounts for the energy reduction due to the extra scattering event:

$$\Phi_M(\tilde{r}(x)) = \rho' \Phi_M(r(x))$$

¹⁰E. d'Eon and G. Irving. “A quantized-diffusion model for rendering translucent materials”. In: *ACM transactions on graphics (TOG)* 30.4 (2011), pp. 1–14.

¹¹C. Grosjean. “A high accuracy approximation for solving multiple scattering problems in infinite homogeneous media”. In: *Il Nuovo Cimento (1955-1965)* 3.6 (1956), pp. 1262–1275.



This expression uses the previous monopole solution , though its diffusion coefficient must be replaced with a non-classical version given by

$$D_G = \frac{2\sigma_a + \sigma'_s}{3(\sigma_a + \sigma'_s)^2}$$

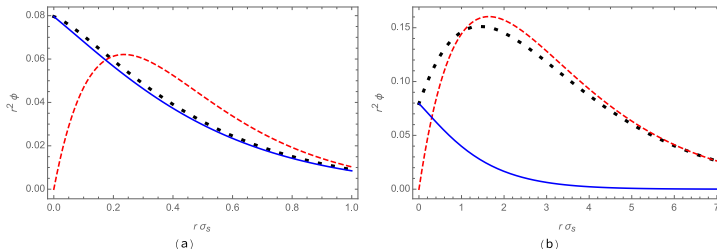


Figure: Comparison of Classical and Non-Classical Diffusion Solutions. The two plots compare the fluence rate due to a point source for (a) a low albedo ($\rho = 1/3$) and (b) a high albedo ($\rho = 0.9$). In both cases, Grosjean's non-classical monopole (blue) is a considerably better match to the exact solution (black) compared to the classical diffusion solution (red).

¹²M. Pharr et al. *Physically based rendering: From theory to implementation*. Morgan Kaufmann, 2016.



The dipole solution



The simplest kind of geometric setup that contains a surface is a semi-infinite half space, i.e., a medium that fills all space below a planar surface of infinite lateral extent. The region above is modeled as a dielectric without any kind of scattering (i.e., a vacuum).

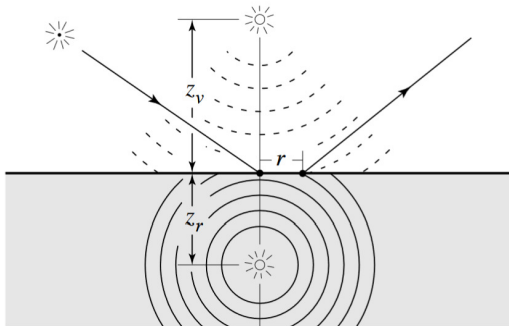


Figure: An incoming ray is transformed into a dipole source for the diffusion approximation.



The dipole solution



The effects of a boundary can be approximated by the **method of images**, where a negative source is placed on the vacuum side of the boundary at position $(0, 0, z_v)^t$ with $z_v < 0$.

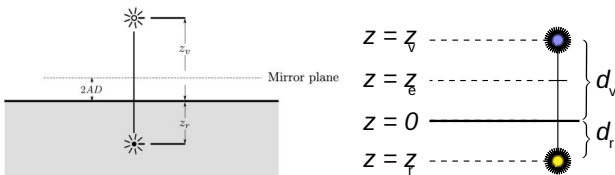


Figure: Basic Setting for the Dipole Approximation to the Solution to the Diffusion Equation. A light source with positive flux is placed at a position $z = z_r$ inside the medium, below the point where incident illumination arrives, and a second light with an equal amount of negative flux is placed at $z = z_v$ above the medium. These sources are placed so that the flux for the two of them cancels out at a height z_e above the boundary, fulfilling the linearized boundary condition. The fluence rate that results from subtracting the closed-form solution for each of them, Equation (15.25), at the medium's boundary is a reasonable approximation of the fluence rate at the boundary due to subsurface scattering.



For simplicity, we assume that the boundary is located at $z = 0$ with a normal $n = (0, 0, -1)^t$ so that positive values on the axis correspond to points inside the medium. Let η denote the relative index of refraction over the boundary.

The *negative contribution* of this 'virtual' light source subtracts a portion of the 'real' light source to account for the combined effect of internal reflections and illumination that has left the medium and can no longer scatter:

$$\Phi_D(r(x)) = \Phi_M(\tilde{d}_r) - \Phi_M(\tilde{d}_v)$$

where $d_r = \sqrt{r^2 + z_r^2}$ and $d_v = \sqrt{r^2 + z_v^2}$ are the straight-line distances from the evaluation point to the real and virtual light source.

We can find a matching vector irradiance value using the monopole solution

$$E_D(r(x)) = \tilde{E}_M(d_r) - \tilde{E}_M(d_v)$$

The z component of this expression is given by

$$-n \cdot E_D(r(x)) = \frac{1}{4\pi} \left[\frac{z_r(1 + d_r\sigma_{tr})}{d_r^3} e^{-\sigma_{tr}d_r} - \frac{z_v(1 + d_r\sigma_{tr})}{d_v^3} e^{-\sigma_{tr}d_v} \right]$$



We assume that the real light source depth z_r is specified, and that the depth z_v of the virtual light source must be set to correct for the boundary conditions.

As we move from the boundary into the region filled by vacuum, it is intuitive that the fluence rate should decrease fairly quickly due to the inverse squared falloff in a region of space that does not scatter.

If we model the fluence rate along the z -axis using a first-order Taylor expansion of the boundary conditions at $z = 0$, then this linear function eventually reaches 0 (and then negative values) at a *linear extrapolation depth* of $z_e < 0$.

Idea: Mirror the real light source across the mirror plane $z = z_e$ defined by this extrapolation depth to obtain the virtual light source depth $z_v = 2z_e - z_r$.

This ensures that the fluence rate of the dipole is exactly equal to 0 at $z = z_e$ above the half-space.



To use the dipole solution in a light transport simulation, we'll need to know how much light actually leaves the surface.

To find the amount of diffusely scattered light leaving at the boundary, we can integrate the internal radiance distribution L_d over the hemisphere:

$$\begin{aligned} E_d(x) &= \int_{\Omega} (1 - F_r(\eta^{-1}, \cos(\theta)) L_d(x, \omega \cos(\theta))) d\omega \\ &= \int_{\Omega} \left(1 - F_r(\eta^{-1}, \cos(\theta)) \left(\frac{1}{\pi} \Phi_D(x) + \frac{3}{4\pi} \omega \cdot E_E(x) \right) \right) d\omega \\ &= \underbrace{\int_{\Omega} (1 - F_r(\eta^{-1}, \cos(\theta))) \frac{1}{\pi} \Phi_D(x) d\omega}_{E_{d, \Phi_D}(x)} \\ &\quad + \underbrace{\int_{\Omega} \left(1 - F_r(\eta^{-1}, \cos(\theta)) \frac{3}{4\pi} \omega \cdot E_E(x) \right) d\omega}_{E_{d, E_D}(x)} \end{aligned}$$



For the fluence-related part we get:

$$\begin{aligned}E_{d,\Phi_D(x)} &= \int_0^{2\pi} \int_0^{\pi/2} (1 - F_r(\eta^{-1}, \cos(\theta))) \frac{1}{4\pi} \Phi_D(x) \cos(\theta) \sin(\theta) d\theta d\phi \\&= \frac{1}{2} \Phi_D(x) \int_0^{\pi/2} (1 - F_r(\eta^{-1}, \cos(\theta))) \cos(\theta) \sin(\theta) d\theta \\&= \Phi_D(x) \left(\frac{1}{4} - \frac{1}{4} \bar{F}_{r,1} \right).\end{aligned}$$

where the i -th moment of the Fresnel reflectance function is given by

$$\bar{F}_{r,i}(\eta) = \int_0^{\pi/2} F_r(\eta, \cos(\theta)) \sin(\theta) \cos(\theta)^i d\theta$$



For the part depending on E_D , we obtain

$$\begin{aligned} E_{d,\Phi_D}(x) &= \int_0^{2\pi} \int_0^{\pi/2} (1 - F_r(\eta^{-1}, \cos(\theta))) \left(\frac{3}{4\pi} \boldsymbol{\omega} \cdot E_D(x) \right) \cos(\theta) \sin(\theta) d\theta d\phi \\ &= \int_0^{\pi/2} (1 - F_r(\eta^{-1}, \cos(\theta))) \frac{3\cos(\theta)}{2} (\mathbf{n} \cdot E_D(x)) \cos(\theta) \sin(\theta) d\theta \\ &= \mathbf{n} \cdot E_D(x) \left(\frac{1}{2} - \frac{3}{2} \bar{F}_{r,2} \right). \end{aligned}$$

Summary: This formula deliver a method of evaluating the radiant exitance at a position x on the boundary due to an internal source at depth z_r .

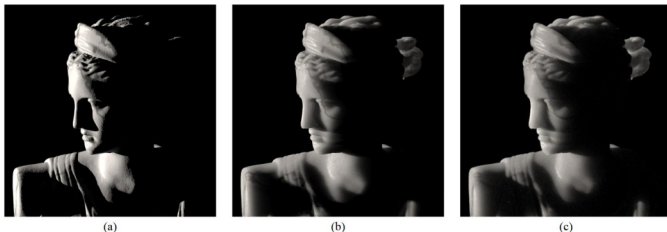
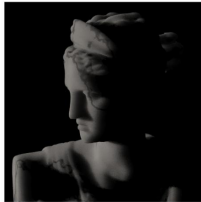


Figure: A simulation of subsurface scattering in a marble bust. The marble bust is illuminated from behind and rendered using: (a) the BRDF approximation (in 2 minutes), (b) the BSSRDF approximation (in 5 minutes), and (c) a full Monte Carlo simulation (in 1250 minutes). Notice how the BSSRDF model matches the appearance of the Monte Carlo simulation, yet is significantly faster.

¹³H. W. Jensen et al. "A practical model for subsurface light transport". en. In: *Proceedings of the 28th annual conference on Computer graphics and interactive techniques - SIGGRAPH '01*. Not Known: ACM Press, 2001, pp. 511–518. URL: <http://portal.acm.org/citation.cfm?doid=383259.383319> (visited on 07/03/2019).



(d)



(e)



(f)

Figure: The images in (d–f) show the different components of the BSSRDF: (d) single scattering term, (e) diffusion term, and (f) Fresnel term.



Figure: Rendering a human face using a single-layer skin model. The classical dipole model (left) is frequency-limited and results in a waxy-looking appearance, particularly on the lips. A multipole model can create very realistic results, but requires additional material parameters which are difficult to measure and unintuitive to edit. Our quantized-diffusion model (right) produces accurate all-frequency subsurface scattering, achieves much of the realism of multilayer models and allows easy appearance editing.



Figure: Applications of quantized-diffusion BSSRDF in production at WETA Digital



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